Self-starting Siphon Priming: Geometry, Dynamics, Losses, and Computation

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20 November 2025

Abstract

Self-starting siphons use a deliberate drop in the centreline to generate a momentum-bearing water slug that climbs a crest without external priming. This article reconnects the full narrative: (i) a physical start-up description with explicit initial conditions, (ii) an exact container volume map for arbitrary internal pipe layouts (piecewise $h_c(L)$) and a constant-area simplification, (iii) modelling limits (cavitation head margin, capillarity, and non-hydrostatic inlet correction), (iv) a detailed filling-phase momentum balance with losses, and (v) an analytic upper bound on the maximum crest height reachable in the idealised lossless, fixed-h_c case, expressed directly in terms of the piecewise-linear geometry. A complete Python integration script is provided to evolve the two-phase ODE system with Churchill friction, minor losses, events, and optional cavitation flagging.

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1 Introduction and mental model

Self-starting siphons differ from conventional inverted-U siphons in that they do not require an externally imposed pressure difference or pre-filling of the tube. Instead, the geometry is arranged so that an initial drop converts gravitational potential energy into kinetic energy in a finite slug of water, and the resulting momentum carries the slug over a crest before friction and gravity bring it to rest. Once the crest has been crossed, the device behaves like a standard siphon and continues to drain until the head between the reservoir and outlet is exhausted.

The aim of this article is to connect this qualitative picture to a one-dimensional slug-flow model that is simple enough for analysis and simulation but rich enough to capture the essential dynamics. We introduce the centreline geometry, the arclength coordinate $L(t)$ and a volume-based description of the container level, derive the basic lossless equation of motion for the filling phase, add distributed and local (minor) losses, and then write down the coupled system that governs the draining phase once the pipe is full. At the level of this slug model, the combined effect of pressure and gravity on the water in the pipe can be summarised very compactly: the net driving force is

$$F_{\text{drive}} = \rho g A\big[h_c(t) - h(L(t))\big], \tag{1}$$

where $h_c(t)$ is the container free-surface elevation and $h(L(t))$ is the centreline height of the water–air front. This driving term is non-zero whenever the front is below the free surface, i.e. while $h(L(t)) < h_c(t)$, so after the initial drop into the lower leg the slug generally re-crosses the container level with finite speed before decelerating towards the crest. By contrast, in a simple inverted-U siphon the driving head vanishes at the crest after the initial drop, so the flow does not enjoy the same momentum advantage.

In practice there are two experimental "wetting" procedures (pre-filling the leg up to a point near B, or dipping the inlet until the interface reaches B) which both lead, at the start of the dynamical filling phase, to the same initial conditions: the front sits at the end of segment AB and is momentarily at rest, while the container level passes through A. We take $t = 0$ at that moment and show below that both procedures collapse to

$$t_0 = 0, \qquad L(0) = L_1, \qquad \dot{L}(0) = 0, \qquad h_c(0) = h_0.$$

Self-starting siphon behaviour and priming have been discussed in the fluid-mechanics and physics-education literature; see, for example, Potter and Barnes [6], Jumper and Stanchev [7],

Zheng et al. [8], Ortloff and Kassinos [9], and Sedlar et al. [10]. The slug-flow model developed below focuses on the transient filling and draining dynamics of a self-starting siphon, including frictional and minor losses.

Throughout, we make the following modelling assumptions:

- The fluid is incompressible and Newtonian with constant density ρ and viscosity μ (e.g. water at room temperature).
- The pipe is rigid, stationary, and has a uniform circular cross-section of diameter D and area A; the centreline geometry may be curved but lies in a vertical plane.
- The container is large enough that the velocity at the free surface is negligible, so the container remains hydrostatic at each instant.
- Both the container free surface and the outlet discharge to the same atmospheric pressure, and the external air dynamics are neglected.

2 Geometry and coordinate systems

We model a self-starting siphon consisting of a rigid container connected to a rigid pipe. Water flows from the container into the pipe under gravity. The geometry is fully described in a two-dimensional vertical plane by five node points and two coordinate systems.

2.1 Global Cartesian coordinate system (x, h)

We introduce a global Cartesian system:

- Horizontal coordinate: x (in metres), increasing to the right.
- Vertical coordinate: h (in metres), increasing upward.

We choose the vertical origin such that $h = 0$ at the container base, so all heights (including $h_0, h_1, \ldots, h_4$) are measured vertically upward from this level. In particular, h_0 is the initial water depth in the container. All heights in the model (container level, pipe elevations) are measured as values of h. The pipe centreline lies entirely in this (x, h) plane.

2.2 Pipe centreline: five node points and four segments

The pipe centreline is defined by five node points

$$A(0, h_0), \quad B(x_1, h_1), \quad C(x_2, h_2), \quad D(x_3, h_3), \quad E(x_4, h_4),$$

connected by straight segments:

$$\begin{aligned}
&\text{Segment 1: from } A \text{ to } B, \text{ length} && L_1 = \sqrt{(x_1 - 0)^2 + (h_1 - h_0)^2}, \\
&\text{Segment 2: from } B \text{ to } C, \text{ length} && L_2 = \sqrt{(x_2 - x_1)^2 + (h_2 - h_1)^2}, \\
&\text{Segment 3: from } C \text{ to } D, \text{ length} && L_3 = \sqrt{(x_3 - x_2)^2 + (h_3 - h_2)^2}, \\
&\text{Segment 4: from } D \text{ to } E, \text{ length} && L_4 = \sqrt{(x_4 - x_3)^2 + (h_4 - h_3)^2}.
\end{aligned}$$

The total pipe length is

$$L_{\max} = L_1 + L_2 + L_3 + L_4.$$

The inlet at the container wall is at node A; the outlet is at node E.

We choose the vertical datum such that at $t = 0$ the container free surface passes through node A. Thus

- the vertical coordinate of A is h_0,
- and the initial container level is $h_c(0) = h_0$.

The outlet elevation is

$$h_{\text{out}} = h(L_{\max}) = h_4.$$

2.3 Centreline elevation as a function of arclength, $h(L)$

We use the same symbol L both for:

- a generic arclength coordinate along the pipe centreline (geometry), and
- the wetted length $L(t)$ (state variable) during the filling phase.

When we write L without an explicit time argument, we mean a generic arclength coordinate measured from node A; when we write $L(t)$, we mean the current wetted length.

As a geometric coordinate, L runs from the inlet to the outlet:

$$L = 0 \text{ at } A, \quad L = L_1 \text{ at } B, \quad L = L_1 + L_2 \text{ at } C, \quad L = L_1 + L_2 + L_3 \text{ at } D, \quad L = L_{\max} \text{ at } E.$$

The elevation of the pipe centreline as a function of arclength is denoted $h(L)$. Since each segment is straight, $h(L)$ is piecewise linear in L:

$$h(L) = \begin{cases} h_0 + \dfrac{L}{L_1}(h_1 - h_0), & 0 \le L \le L_1 \text{ (segment } A\text{–}B), \\ h_1 + \dfrac{L - L_1}{L_2}(h_2 - h_1), & L_1 \le L \le L_1 + L_2 \text{ (segment } B\text{–}C), \\ h_2 + \dfrac{L - L_1 - L_2}{L_3}(h_3 - h_2), & L_1 + L_2 \le L \le L_1 + L_2 + L_3 \text{ (segment } C\text{–}D), \\ h_3 + \dfrac{L - L_1 - L_2 - L_3}{L_4}(h_4 - h_3), & L_1 + L_2 + L_3 \le L \le L_{\max} \text{ (segment } D\text{–}E). \end{cases}$$

This function $h(L)$ encodes the entire geometry of the pipe (drops, rises, and crest). For later analytic work it is convenient to introduce cumulative lengths

$$S_0 = 0, \quad S_1 = L_1, \quad S_2 = L_1 + L_2, \quad S_3 = L_1 + L_2 + L_3, \quad S_4 = L_{\max},$$

and to rewrite $h(L)$ on each segment in affine form

$$h(L) = c_i + s_i L, \qquad s_i = \frac{h_i - h_{i-1}}{L_i}, \qquad c_i = h_{i-1} - s_i S_{i-1}, \tag{2}$$

for $L \in [S_{i-1}, S_i]$. If segment i makes an angle θ_i with the horizontal then its slope s_i is simply $s_i = \sin\theta_i$, so only the vertical projection of each leg enters the priming analysis.

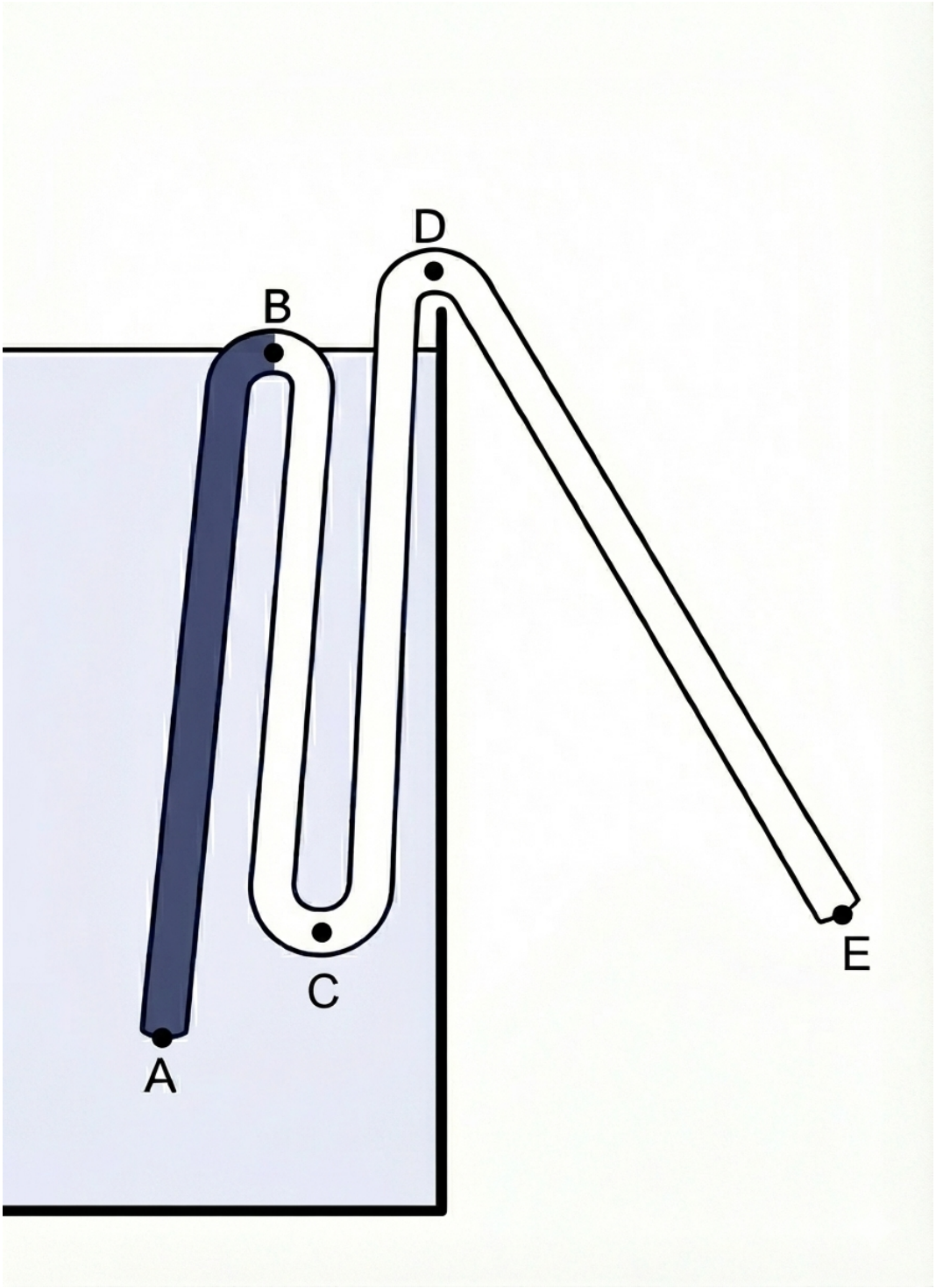


Figure 1: Schematic diagram of the self-starting siphon geometry showing the container, pipe centreline with five node points (A, B, C, D, E), and the coordinate system. The initial water level h_0 is at node A, and the pipe follows a path designed to create a momentum-driven slug that can climb over the crest at D before draining through the outlet at E.

2.4 Wetted length coordinate $L(t)$ and the advancing front

During the filling phase the pipe is not yet completely full. We introduce the wetted length:

- $L(t)$ is the length of pipe actually filled with water at time t,
- measured along the centreline from node A,
- with $0 < L(t) \leq L_{\max}$.

The water–air interface (front) is located at the arclength position $L(t)$, and its elevation is

$$h_{\text{front}}(t) = h(L(t)).$$

When the front has advanced all the way to the outlet, $L(t) = L_{\max}$ and the pipe becomes completely full.

3 Container, volume map, and effective area

3.1 Container and pipe properties

We consider a rigid vertical cylindrical container of inner radius R_c. Its geometric cross-sectional area is

$$A_{\text{cyl}} = \pi R_c^2.$$

In the model we do not work directly with A_{cyl}, but with an effective free-surface area A_0 defined as the cross-sectional area actually occupied by water in the container, after subtracting any internal solid obstructions (such as the pipe wall). In other words, A_0 is the net area of the water region in a horizontal cross-section, not including the pipe interior.

- *Container free-surface area (assumed constant):*

  $$A_0 \ [\text{m}^2].$$

  A_0 is the effective area of the water region in the cylindrical container, i.e. the cylindrical cross-section minus the cross-sectional area of any internal obstructions such as the pipe wall. The interior of the pipe (whether filled with water or air) is not part of the container control volume. With this definition, the water volume in the container is

  $$V_{\text{cont}}(t) = A_0 h_c(t),$$

  where $h_c(t)$ is the container free-surface elevation measured against the cylinder wall.

- *Pipe cross-sectional area:*

  $$A \ [\text{m}^2],$$

  with circular diameter D. The water volume inside the pipe at time t is

  $$V_{\text{pipe}}(t) = AL(t).$$

By definition, any portion of the pipe interior, even if it lies below the current free surface and is geometrically inside the cylindrical shell, is counted as part of the pipe volume V_{pipe}, not the container volume V_{cont}. The displacement effect of the pipe wall on the free surface is already built into the choice of A_0 as an effective area, so $V_{\text{cont}} = A_0 h_c$ and $V_{\text{pipe}} = AL(t)$ together account for all the water in the system without double-counting.

3.2 On the effective area A_0 for complex internal pipe layouts

The physical prototype motivating this model has a meandering pipe inside the container: a horizontal plane at a given elevation may intersect zero, one, or several legs of the pipe. In such a configuration, the area occupied by water in a horizontal cross-section is not literally "cylinder area minus one pipe" at all heights, because the internal legs can be tilted rather than strictly vertical.

To make this precise, consider a horizontal slice at elevation η. Let We denote by $n(\eta)$ the number of individual pipe legs intersected by that slice. On each leg i the pipe centreline has slope s_i in the (x, h) plane, with $s_i = (h_i - h_{i-1})/L_i = \sin\theta_i$ as in (2). A horizontal slice cuts each cylindrical leg along an oblique ellipse whose area is larger than the true circular cross-section A by a factor that depends on the tilt angle. For a straight leg of infinite extent, elementary geometry gives a horizontal footprint

$$A_{\text{foot},i} = \frac{A}{|s_i|}$$

provided the leg is not exactly horizontal ($s_i \neq 0$). In our prototype the internal legs are all noticeably tilted, so this expression is a good approximation over the range of heights spanned by the container.

With these definitions, the effective water-bearing area in the slice is

$$A_{\text{eff}}(\eta) = A_{\text{cyl}} - \sum_i n_i(\eta)\, A_{\text{foot},i},$$

where $n_i(\eta)$ is 1 if the slice intersects leg i and 0 otherwise. The exact container water volume (excluding the pipe interior) is then

$$V_{\text{cont}}(t) = \int_0^{h_c(t)} A_{\text{eff}}(\eta)\, d\eta.$$

During the filling phase the total water volume is conserved, so

$$V_{\text{cont}}(t) + V_{\text{pipe}}(t) = V_{\text{cont}}(0) + V_{\text{pipe}}(0).$$

Introducing

$$F(h) = \int_0^{h} A_{\text{eff}}(\eta)\, d\eta,$$

we can write

$$F(h_c(t)) + AL(t) = F(h_0) + AL_1.$$

Equivalently, expressing the container level as a function of the wetted length,

$$F(h_c(L)) = F(h_0) + A(L_1 - L).$$

In the fully general case, $h_c(L)$ is therefore obtained from an implicit relation involving F and its inverse:

$$h_c(L) = F^{-1}\Big(F(h_0) + A(L_1 - L)\Big).$$

For a meandering pipe with straight legs, $A_{\text{eff}}(\eta)$ and hence $F(h)$ are piecewise constant/linear functions of h, so $h_c(L)$ is piecewise linear with slope changes at elevations where the number $n(\eta)$ of intersected legs changes.

In this paper, to keep the analytical model compact and highlight the main dynamical mechanisms, we adopt the simpler relation

$$V_{\text{cont}}(t) = A_0 h_c(t),$$

with A_0 treated as a constant. This corresponds to either:

1. an idealised container geometry in which the pipe occupies a shaft of constant footprint across the range of h_c, or

2. a slender-pipe approximation for the real geometry, in which the pipe diameter is small compared to the container radius and the variation of $n(\eta)$ with height only produces a small perturbation in the free-surface height.

Under either interpretation, taking A_0 as constant leads to the linear volume–height relation (3) used below and greatly simplifies the coupling between the container level and the advancing front, while retaining a faithful description of the dominant siphon dynamics. For a given internal layout, the function $A_{\text{eff}}(\eta)$ and its integral $F(h) = \int_0^h A_{\text{eff}}(\eta)\, d\eta$ provide an exact (though piecewise-defined) map from container level to container volume; the constant-A_0 model is a compact specialisation that captures the leading-order behaviour.

3.3 Volume conservation in the filling phase

3.3.1 Global volume balance (container + pipe)

During the filling phase, no water has yet left the outlet, so the total volume of water in the combined container–pipe system remains constant:

$$V_{\text{tot}} = V_{\text{cont}}(t) + V_{\text{pipe}}(t) = \text{constant}.$$

Using

$$V_{\text{cont}}(t) = A_0 h_c(t), \qquad V_{\text{pipe}}(t) = AL(t),$$

we obtain

$$A_0 h_c(t) + AL(t) = V_{\text{tot}}.$$

At $t = 0$ we already found $V_{\text{tot}} = A_0 h_0 + AL_1$. Therefore, for any later time t in the filling phase,

$$A_0 h_c(t) + AL(t) = A_0 h_0 + AL_1. \tag{3}$$

Equation (3) is the global volume conservation relation for the filling phase.

3.3.2 Container level as a function of wetted length, $h_c(L)$

Solving the conservation relation (3) for h_c in terms of L gives

$$A_0 h_c(L) = A_0 h_0 + AL_1 - AL,$$

so that

$$h_c(L) = h_0 - \frac{A}{A_0}\,(L - L_1).$$

Thus during the filling phase,

$$h_c(L) = h_0 - \frac{A}{A_0}(L - L_1). \tag{4}$$

This is the key relation: the container free-surface elevation expressed directly as a function of the wetted length, obtained from plain volume conservation.

Important observations:

- $h_c(L)$ decreases linearly as L grows.

- The volume of water occupying any part of the pipe is already accounted for via the term AL. One does not need a separate "pipe volume under the surface" term; it is automatically included in $V_{\text{pipe}} = AL$.
- In the filling phase, $h_c(t)$ is not an independent state: it is fully determined by $L(t)$.

4 Validity limits and missing physics

The slug model developed below is intentionally minimal. It remains accurate only while certain physical assumptions hold. This section collects those limits, with enough explanation that a reader familiar with basic mechanics and Bernoulli's law can see where the model may break down.

- **Hydrostatic inlet assumption.** The inlet pressure is taken as $p(0,t) = p_{\text{atm}} + \rho g\big(h_c(t) - h_0\big)$, i.e. the container is assumed to remain hydrostatic at each instant: pressure at a given depth depends only on the instantaneous height of the free surface, not on how fast it is moving. This is valid when the free surface moves slowly compared with the natural acceleration scale of the system,

$$|\dot{h}_c| \ll \sqrt{gD}, \qquad |\ddot{h}_c| \ll g,$$

  so that unsteady and dynamic effects at the inlet are small. The first inequality says, very loosely, that the container level should not drop by an amount comparable to the pipe diameter over the time it takes a gravity wave to cross that diameter. If these conditions are violated, the inlet behaves more like a rapidly accelerating jet; in that case an unsteady Bernoulli term can be interpreted as an additional effective head of order

$$\Delta h_{\text{unsteady}} \approx -\frac{A}{A_0}\,\frac{\dot{h}_c\,\dot{L}}{g},$$

  which lies beyond the present model. In the experiments that motivated this work the inequalities above are comfortably satisfied, so the hydrostatic approximation is a good one.

- **Cavitation / tensile failure.** Along the centreline, and especially near the crest, the absolute pressure must remain above vapour pressure (or the tensile strength of the liquid) to avoid column separation. In terms of "head", this means that the vertical distance between the container surface and the crest, after subtracting any friction/loss head already spent, must not be so large that the pressure at the crest would have to fall below vapour pressure. A conservative requirement is that the available head between container and crest, minus the loss head to that point, exceeds the vapour head,

$$h_c - h_{\text{crest}} - h_{\text{loss,crest}} \geq h_{\text{vap}}, \qquad h_{\text{vap}} = \frac{p_{\text{atm}} - p_{\text{vap}}}{\rho g}. \tag{5}$$

  If (5) is violated, the modelled negative pressures are not sustainable: in reality small vapour bubbles appear (cavitation) or air is pulled in, and the flow no longer behaves as a single continuous slug of water. Modern discussions of the siphon height limit and negative pressures include Boatwright et al. [12] and Vera et al. [13]; the role of water under tension in plant xylem is reviewed by Tyree [14].

- **Capillarity / surface tension.** For very small diameters D the meniscus at the inlet can sustain a finite pressure jump. A simple estimate is the Jurin-like height

$$h_\sigma \approx \frac{4\sigma\cos\theta}{\rho g D},$$

where σ is the surface tension and θ the contact angle. For millimetre-scale pipes this height can be a few centimetres, which is small compared with the overall siphon height but can matter for very gentle start-up. A net head at least this large is required to break the meniscus and start the motion; once a finite velocity is established, the standard loss model applies.

- **Slug-model breakdown.** The present model treats the water as a single, one-dimensional slug with a sharp front: the model assumes almost-uniform velocity across each cross-section and a single connected water column. Strong three-dimensional curvature, vigorous turbulence, air entrainment, or repeated column separation can all invalidate this picture. In such regimes a two-phase or fully resolved CFD approach is more appropriate, and the present equations should only be used for qualitative guidance.

5 Initial conditions and comparison with an inverted-U siphon

We choose $t = 0$ at the start of the filling phase, when the dynamical motion of the water slug begins. In practice there are two simple wetting procedures that prepare this state:

1. *Pre-fill to B.* The container is slowly filled until the free surface reaches point B on the first leg, fully wetting segment AB. During this quasi-static process the wetted length simply tracks the container level; at the moment the surface passes B one has $L = L_1$.

2. *Dip to B.* The inlet is dipped into a reservoir so that the water rises inside the leg until the front reaches B, after which the leg is placed in its operating position. Again, the filling up to B is slow compared with the subsequent motion, so the front arrives at B with negligible speed.

In both cases the slow preparatory motion ends with the same configuration: the front is at the end of segment AB, the container surface passes through A at height h_0, and the slug is momentarily at rest. It is therefore natural, and physically realistic, to adopt the initial conditions

$$t_0 = 0, \qquad h_c(0) = h_0, \qquad L(0) = L_1, \qquad \dot{L}(0) = 0. \tag{6}$$

The total water volume in the container + pipe system at $t = 0$ is

$$V_{\text{tot}} = V_{\text{cont}}(0) + V_{\text{pipe}}(0) = A_0 h_0 + A L_1.$$

This quantity V_{tot} will be crucial in the volume-based derivation of $h_c(L)$ during the filling phase; see section 3.

It is useful to contrast this self-starting configuration with a simple inverted-U siphon. In the inverted-U case the driving head (difference in height between the reservoir surface and the outlet) is already fully "spent" when the flow reaches the crest, so the fluid has no further pressure advantage once it passes the highest point. In the present geometry, by contrast, the driving term $\rho g A\big[h_c(t) - h(L(t))\big]$ remains non-zero as long as the front elevation $h(L(t))$ lies below the container free surface $h_c(t)$. After the initial drop from A into the lower leg, the slug therefore tends to re-cross the container level with finite speed before decelerating towards the crest, which is the key difference underlying the self-priming behaviour.

6 Dynamics in the filling phase

In the filling phase the pipe is only partially full: the front is at position $L(t)$ and the outlet at E is still dry. The water in the pipe is modelled as a one-dimensional slug of length $L(t)$, cross-sectional area A, and uniform mean velocity

$$v(t) = \dot{L}(t).$$

Its mass is

$$m(t) = \rho A L(t),$$

where ρ is the fluid density.

The driving force comes from the instantaneous head difference between the container free surface at height $h_c(t)$ and the front elevation $h(L(t))$, while wall friction and local disturbances in the pipe generate resistive forces. In this section we first derive the lossless equation of motion from momentum conservation and then incorporate the loss models.

6.1 Lossless momentum balance

In the absence of friction and minor losses, the net driving force on the slug is

$$F_{\text{drive}}(t) = \rho g A \big[h_c(t) - h(L(t))\big],$$

where g is the gravitational acceleration. This result is derived below by integrating the pressure and gravity contributions along the pipe.

Newton's second law in the direction of motion reads

$$m(t)\,\ddot{L}(t) = F_{\text{drive}}(t).$$

Substituting $m(t) = \rho A L(t)$ and cancelling ρA yields

$$L(t)\,\ddot{L}(t) = g\big[h_c(t) - h(L(t))\big].$$

During the filling phase the container level $h_c(t)$ is not an independent state variable: it is determined by global volume conservation, and we may regard h_c as a function of the wetted length via (4). Using this relation, the lossless filling-phase equation of motion can be written in closed form as

$$\ddot{L} = \frac{g}{L}\Big[h_0 - \frac{A}{A_0}(L - L_1) - h(L)\Big]. \tag{7}$$

Equation (7) is integrated from the initial conditions

$$L(0) = L_1, \qquad \dot{L}(0) = v_0,$$

up to the time t^* at which the front first reaches the outlet, $L(t^*) = L_{\max}$.

6.2 Detailed derivation of the net driving force

We now show how the expression

$$F_{\text{drive}} = \rho g A \big[h_c(t) - h(L)\big]$$

arises from pressure and gravity acting on the water in the pipe.

Introduce a local coordinate ξ along the pipe centreline, measured from the inlet, so that ξ runs from 0 at the entrance to $L(t)$ at the front. At position ξ the pipe centreline is at height $h(\xi)$, the pressure is $p(\xi, t)$, and the tangent to the pipe makes some angle to the horizontal. Consider a thin fluid slice between ξ and $\xi + d\xi$:

- cross-sectional area: A,
- thickness: $d\xi$,
- volume: $A\,d\xi$,
- mass: $dm = \rho A\,d\xi$.

Forces tangential to the pipe on this slice are:

- pressure force from the upstream side: $p(\xi, t)\,A$,
- pressure force from the downstream side: $-p(\xi + d\xi, t)\,A$,
- component of the weight along the pipe: $-\rho g A\,\frac{dh}{d\xi}\,d\xi$,

where the sign convention is chosen so that increasing ξ follows the direction of motion and $dh/d\xi$ accounts for the local inclination of the pipe. The total tangential force on the slice is therefore

$$dF = -A\frac{\partial p}{\partial \xi}\,d\xi - \rho g A\,\frac{dh}{d\xi}\,d\xi.$$

Integrating from $\xi = 0$ to $\xi = L(t)$ gives the net force on the whole slug:

$$F_{\text{net}} = \int_0^L dF = -A\int_0^L \frac{\partial p}{\partial \xi}\,d\xi - \rho g A\int_0^L \frac{dh}{d\xi}\,d\xi.$$

The second integral is purely geometric:

$$\int_0^L \frac{dh}{d\xi}\,d\xi = h(L) - h(0),$$

so

$$-\rho g A\int_0^L \frac{dh}{d\xi}\,d\xi = \rho g A\big[h(0) - h(L)\big].$$

For the pressure term we obtain

$$-A\int_0^L \frac{\partial p}{\partial \xi}\,d\xi = -A\big[p(L,t) - p(0,t)\big] = A\big[p(0,t) - p(L,t)\big].$$

Combining the two contributions yields

$$F_{\text{net}} = A\big[p(0,t) - p(L,t)\big] + \rho g A\big[h(0) - h(L)\big].$$

We now relate the end pressures $p(0,t)$ and $p(L,t)$ to the container free surface and the outlet:

- At the free surface in the container the pressure is p_{atm}. Hydrostatic equilibrium in the container gives the pressure at the pipe entrance,

$$p(0,t) = p_{\text{atm}} + \rho g\big[h_c(t) - h(0)\big].$$

- At the outlet the jet discharges to the atmosphere at height $h(L)$, so

$$p(L,t) = p_{\text{atm}}.$$

Substituting into the expression for F_{net} and cancelling p_{atm},

$$F_{\text{net}} = \rho g A\big[h_c(t) - h(0)\big] + \rho g A\big[h(0) - h(L)\big] = \rho g A\big[h_c(t) - h(L)\big].$$

Thus the combined effect of the hydrostatic pressure in the container and the weight of the water in the pipe is equivalent to a single driving head between the container free surface and the front. In the lossless case $F_{\text{drive}} = F_{\text{net}}$, and inserting this into Newton's law recovers the lossless equation of motion (7).

6.3 Loss models: distributed friction and local (minor) losses

In reality the pipe walls and geometric disturbances dissipate energy. We model these effects using standard head-loss expressions.

Distributed wall friction (Darcy–Weisbach)

The head loss due to viscous friction along a length L of straight pipe with diameter D and mean velocity v is written in Darcy–Weisbach form as

$$h_f = f \frac{L}{D} \frac{v^2}{2g},$$

where f is the Darcy friction factor, which depends on the Reynolds number

$$\mathrm{Re} = \frac{\rho v D}{\mu}$$

and the relative roughness of the pipe. For laminar flow ($\mathrm{Re} \lesssim 2000$) one has

$$f = \frac{64}{\mathrm{Re}},$$

while for turbulent and transitional flow an explicit all-regime correlation due to Churchill [2] is particularly convenient. In the form given by Churchill [2] it reads

$$\frac{f}{8} = \left[\left(\frac{8}{\mathrm{Re}} \right)^{12} + \frac{1}{(A_1 + A_2)^{3/2}} \right]^{1/12}, \tag{8}$$

$$A_1 = \left[-2.457 \ln \left(\left(\frac{7}{\mathrm{Re}} \right)^{0.9} + 0.27 \frac{\varepsilon}{D} \right) \right]^{16}, \tag{9}$$

$$A_2 = \left(\frac{37530}{\mathrm{Re}} \right)^{16}, \tag{10}$$

where ε/D is the relative roughness. Equations (8)–(10) are used to evaluate the friction factor f in the loss models below. For comparison or validation one may still refer to the implicit Colebrook–White equation or a Moody diagram [3, 4]. In many siphon applications a representative constant value of f (based on the typical operating Re) is adequate.

Local (minor) losses

Bends, entrances, exits, contractions, valves, and other fittings are represented by local (or "minor") losses of the form

$$h_m = K \frac{v^2}{2g},$$

where K is a dimensionless loss coefficient determined by the fitting geometry. If several elements are present along the wetted portion of the pipe, their effects are additive:

$$h_{\mathrm{minor,tot}} = \left(\sum_i K_i \right) \frac{v^2}{2g}.$$

Published tables give typical K_i values for common components (sharp or rounded entrances, elbows, valves, sudden expansions, and so on), and compilations such as Crane [5] are widely used in engineering practice.

Total head loss and resistive force

Over a wetted length L with mean velocity v the total head loss is modelled as

$$h_{\text{loss}}(v, L) = f\,\frac{L}{D}\,\frac{v^2}{2g} + \Big(\sum_i K_i\Big)\,\frac{v^2}{2g}.$$

In the slug model this head corresponds to a resistive force opposing the motion,

$$F_{\text{loss}} = \rho g A\, h_{\text{loss}}(v, L)\ \text{sgn}(v).$$

Using $v = \dot{L}$ in the filling phase and writing $|v|v = v\ \text{sgn}(v)\, v$ gives

$$F_{\text{loss}} = \frac{\rho A}{2}\Big(f\,\frac{L}{D} + \sum_i K_i\Big)\,|\dot{L}|\,\dot{L}.$$

6.4 Extended equation of motion in the filling phase

Including both the driving force and the resistive force, Newton's second law for the slug becomes

$$\rho A L\,\ddot{L} = \rho g A\big[h_c(t) - h(L)\big] - \frac{\rho A}{2}\Big(f\,\frac{L}{D} + \sum_i K_i\Big)\,|\dot{L}|\,\dot{L}. \tag{11}$$

Cancelling ρA and again using $h_c(L)$ from above, we obtain the extended filling-phase ODE

$$\ddot{L} = \frac{g}{L}\Big[h_0 - \frac{A}{A_0}(L - L_1) - h(L)\Big] - \frac{1}{2L}\Big(f\,\frac{L}{D} + \sum_i K_i\Big)\,|\dot{L}|\,\dot{L}. \tag{12}$$

Equation (12) combines geometry, container drawdown, and both distributed and local losses into a single nonlinear ODE for the front position $L(t)$ during the filling phase. The initial conditions at $t = 0$ are

$$L(0) = L_1, \qquad \dot{L}(0) = v_0,$$

with $h_c(L)$ given by the volume-conservation relation above. The filling phase ends when $L(t)$ first reaches $L_{\max}$.

7 Idealised climb bound and analytical stall height

7.1 Idealised lossless priming model and upper bound

For design purposes it is useful to consider an idealised upper-bound model for the priming process. In this model we make two additional simplifications:

- the container level is held fixed at $h_c(t) \equiv h_0$ (no drawdown), and
- all losses are neglected, $f = 0$ and $\sum_i K_i = 0$.

We retain the same initial conditions as in the full model,

$$L(0) = L_1, \qquad \dot{L}(0) = 0,$$

so that the front starts from rest at the end of the first segment.

Under these assumptions the filling-phase equation reduces to

$$\ddot{L} = \frac{g}{L}\big[h_0 - h(L)\big]. \tag{13}$$

Introducing $v(L) = \dot{L}$ and viewing the velocity as a function of position along the pipe, we can rewrite the second derivative using the chain rule,

$$\ddot{L} = \frac{dv}{dt} = \frac{dv}{dL}\,\frac{dL}{dt} = v\,\frac{dv}{dL},$$

which converts the time derivative into a derivative with respect to L. Substituting this into (13) gives

$$v\,\frac{dv}{dL} = \frac{g}{L}\big[h_0 - h(L)\big]. \tag{14}$$

Integrating from L_1 (where $v = 0$) to a general L yields

$$\frac{1}{2}\big[v(L)^2 - 0\big] = g\int_{L_1}^{L} \frac{h_0 - h(\xi)}{\xi}\,d\xi, \tag{15}$$

so that

$$v(L)^2 = 2g\int_{L_1}^{L} \frac{h_0 - h(\xi)}{\xi}\,d\xi. \tag{16}$$

The integrand has a simple interpretation: $h_0 - h(\xi)$ is the local "excess head" above the front, and the factor $1/\xi$ weights this head by the inverse distance travelled so far. As the front climbs, regions where $h(\xi)$ lies well below h_0 add positive contributions to the integral (accelerating the slug), while regions close to or above h_0 contribute little or even decelerate it.

Closed form for piecewise-linear $h(L)$

On each geometric segment i the elevation can be written in the affine form (2), $h(L) = c_i + s_i L$ for $L \in [S_{i-1}, S_i]$. The contribution of a subinterval $[a, b] \subseteq [S_{i-1}, S_i]$ to the integral in (16) is then

$$\begin{aligned}\int_a^b \frac{h_0 - h(\xi)}{\xi}\,d\xi &= \int_a^b \frac{h_0 - c_i - s_i\xi}{\xi}\,d\xi \\ &= (h_0 - c_i)\int_a^b \frac{d\xi}{\xi} - s_i\int_a^b d\xi \\ &= (h_0 - c_i)\ln\!\left(\frac{b}{a}\right) - s_i(b - a).\end{aligned} \tag{17}$$

It is convenient to denote this segment contribution by

$$\Phi_i(a, b) = (h_0 - c_i)\ln\!\left(\frac{b}{a}\right) - s_i(b - a). \tag{18}$$

For a given candidate L lying on segment k, i.e. $L \in [S_{k-1}, S_k]$, the integral in (16) splits into full segments and a partial segment,

$$\int_{L_1}^{L} \frac{h_0 - h(\xi)}{\xi}\,d\xi = \sum_{i=2}^{k-1} \Phi_i(S_{i-1}, S_i) + \Phi_k(S_{k-1}, L). \tag{19}$$

The stall condition (21) on that segment can thus be written as

$$\sum_{i=2}^{k-1} \Phi_i(S_{i-1}, S_i) + \Phi_k(S_{k-1}, L) = 0, \tag{20}$$

which is a monotone equation for L on $[S_{k-1}, S_k]$ and can be solved with a simple root-finding method such as bisection or a bracketed Newton iteration. The resulting L_{stall} gives the idealised maximum climb $h_{\max} = h(L_{\text{stall}})$ for the chosen geometry. In this ideal model the front comes to rest at the first arclength $L = L_{\text{stall}}$ such that $v(L_{\text{stall}}) = 0$, i.e.

$$\int_{L_1}^{L_{\text{stall}}} \frac{h_0 - h(\xi)}{\xi}\, d\xi = 0. \tag{21}$$

The corresponding maximum centreline elevation achievable by the priming slug in the ideal lossless model is

$$h_{\max} = h\big(L_{\text{stall}}\big). \tag{22}$$

Because this model ignores both drawdown and dissipation, $h_{\max}$ is an upper bound on what can be attained in the full Churchill+loss model for the same geometry and initial configuration.

7.2 Geometric priming criterion

Let L_D denote the arclength coordinate of the crest point D; by construction its elevation is h_3. The idealised lossless model above leads to a simple geometric necessary condition for self-priming:

$$h_{\max} > h_3. \tag{23}$$

If $h_{\max} \le h_3$, then even the most optimistic model (no drawdown, no losses, front starting from rest at L_1) predicts that the front stalls below the crest. In that case the given geometry cannot self-prime, regardless of the detailed friction or wetting conditions. If $h_{\max} > h_3$, the geometry passes this necessary condition; actual self-priming then depends on the full dynamics with continuous Churchill friction and finite container drawdown.

8 Dynamics in the draining phase

Once the front reaches the outlet, the pipe fills completely and the system transitions to a draining phase. The wetted length is then fixed at

$$L(t) = L_{\max}, \qquad t \ge t^*,$$

where t^* is the time of first filling, but the mean velocity $v(t)$ and the container level $h_c(t)$ continue to evolve.

8.1 Container mass balance

In the draining phase the volumetric flow rate through the pipe is

$$Q(t) = Av(t),$$

and this flow rate leaves the container. The container volume therefore satisfies

$$\frac{dV_{\text{cont}}}{dt} = -Q(t) = -Av(t).$$

Using $V_{\text{cont}}(t) = A_0 h_c(t)$ and dividing by A_0 gives

$$\dot{h}_c(t) = -\frac{A}{A_0}\, v(t), \qquad t \geq t^*. \tag{24}$$

Unlike in the filling phase, the total water volume in container plus pipe is no longer constant, so there is no algebraic relation of the form $h_c(L)$.

8.2 Momentum balance with losses in the draining phase

For a completely filled pipe the slug length is $L_{\max}$, the outlet elevation is the fixed height

$$h_{\text{out}} = h(L_{\max}),$$

and the mean velocity $v(t)$ is the dynamical variable.

The momentum balance along the pipe now reads

$$\rho A L_{\max}\, \dot{v} = \rho g A \big[h_c(t) - h_{\text{out}}\big] - \rho g A\, h_{\text{loss}}(v, L_{\max})\ \operatorname{sgn}(v),$$

with

$$h_{\text{loss}}(v, L_{\max}) = \Big(f\,\frac{L_{\max}}{D} + \sum_i K_i\Big)\,\frac{v^2}{2g}.$$

Cancelling ρA yields the draining-phase velocity equation

$$L_{\max}\, \dot{v} = g\big[h_c(t) - h_{\text{out}}\big] - \frac{1}{2}\Big(f\,\frac{L_{\max}}{D} + \sum_i K_i\Big)\,|v|\, v. \tag{25}$$

Equivalently, we may write

$$\dot{v} = \frac{g}{L_{\max}}\big[h_c - h_{\text{out}}\big] - \frac{1}{2L_{\max}}\Big(f\,\frac{L_{\max}}{D} + \sum_i K_i\Big)\,|v|\, v. \tag{26}$$

Together, the equations (24) and (26) form a coupled system for $h_c(t)$ and $v(t)$ in the draining phase.

The initial conditions at the start of draining are taken from the end of the filling phase:

$$v(t^*) = \dot{L}(t^*), \qquad h_c(t^*) = h_0 - \frac{A}{A_0}(L_{\max} - L_1).$$

9 Model summary

The complete model for a self-starting siphon based on the present geometry and volume description can be summarised as follows; it links the centreline geometry $h(L)$ introduced in section 2 to the dynamical variables $L(t)$, $v(t)$, and $h_c(t)$.

1. **Filling phase ($0 \leq t < t^*$).**
   - State variables: wetted length $L(t)$ and its time derivative $\dot{L}(t)$.
   - Container level is constrained by volume conservation:

$$h_c(L) = h_0 - \frac{A}{A_0}(L - L_1).$$

- Dynamics: the front position $L(t)$ evolves according to the nonlinear ODE given in (12), with initial data $L(0) = L_1$ and $\dot{L}(0) = 0$.
- The filling phase ends at the first time t^* such that $L(t^*) = L_{\max}$.

2. **Draining phase ($t \geq t^*$).**

- State variables: mean pipe velocity $v(t)$ and container level $h_c(t)$.
- Kinematics: the pipe remains full ($L = L_{\max}$) and the container height obeys the mass-balance equation (24).
- Dynamics: the velocity satisfies the momentum equation (26), with initial conditions $v(t^*) = \dot{L}(t^*)$ and $h_c(t^*) = h_0 - \frac{A}{A_0}(L_{\max} - L_1)$.

3. **Loss models.**

- Distributed wall friction is modelled via the Darcy–Weisbach relation with friction factor f determined from Reynolds number and pipe roughness.
- Local geometric disturbances are represented by minor-loss coefficients K_i.
- In both phases, distributed and local losses enter the dynamics through the combination

$$f\,\frac{L}{D} + \sum_i K_i,$$

evaluated with $L = L(t)$ in the filling phase and $L = L_{\max}$ in the draining phase.

These equations provide a unified, self-consistent description of the priming and draining dynamics for a self-starting siphon, explicitly connecting the geometry $h(L)$, container drawdown $h_c(L)$, and frictional and minor losses to the evolution of $L(t)$, $v(t)$, and $h_c(t)$ in time.

Key equations

Label	Equation	Meaning
(3)	$A_0 h_c + AL = A_0 h_0 + AL_1$	Global volume conservation during filling (container + pipe).
(4)	$h_c(L) = h_0 - \dfrac{A}{A_0}(L - L_1)$	Container level as function of wetted length.
(1)	$F_{\text{drive}} = \rho g A\big[h_c(t) - h(L)\big]$	Net driving force from pressure and gravity.
(12)	$\ddot{L} = \dfrac{g}{L}\Big[h_0 - \dfrac{A}{A_0}(L - L_1) - h(L)\Big] - \dfrac{1}{2L}\Big(f\,\dfrac{L}{D} + \sum_i K_i\Big)\lvert\dot{L}\rvert\,\dot{L}$	Full filling-phase ODE with drawdown and losses.
(16)	$v(L)^2 = 2g\displaystyle\int_{L_1}^{L} \dfrac{h_0 - h(\xi)}{\xi}\,d\xi$	Ideal lossless velocity as function of position.
(20)–(22)	$\sum_{i=2}^{k-1} \Phi_i(S_{i-1}, S_i) + \Phi_k(S_{k-1}, L) = 0,\ h_{\max} = h(L_{\text{stall}})$	Piecewise-analytic stall condition.
(24)	$\dot{h}_c = -\dfrac{A}{A_0} v$	Mass balance for container in draining.
(26)	$\dot{v} = \dfrac{g}{L_{\max}}(h_c - h_{\text{out}}) - \dfrac{1}{2L_{\max}}\Big(f\,\dfrac{L_{\max}}{D} + \sum_i K_i\Big)\lvert v\rvert\,v$	Momentum balance with losses in draining.
(5)	$h_c - h_{\text{crest}} - h_{\text{loss,crest}} \geq h_{\text{vap}}$	Cavitation head margin at crest.

Variables and parameters

Symbol	Units	Description
$L(t)$	m	Wetted length of the slug along the pipe centreline, measured from A to the front.
$h(L)$	m	Centreline elevation at arclength L; piecewise linear according to the geometry.
$h_c(t)$	m	Container free-surface elevation measured from the container base.
h_0	m	Initial container height $h_c(0)$ and vertical coordinate of point A.
$v(t)$	$\mathrm{m\,s^{-1}}$	Mean slug velocity $v = \dot{L}$ in filling; mean pipe velocity in draining.
A	$\mathrm{m^2}$	Pipe cross-sectional area for circular diameter D, $A = \pi D^2/4$.
A_0	$\mathrm{m^2}$	Effective container cross-sectional area in the constant-area model.
L_i	m	Length of segment i of the pipe; $L_{\max} = L_1 + L_2 + L_3 + L_4$.
ρ, μ	$\mathrm{kg\,m^{-3}}$, Pa s	Fluid density and dynamic viscosity.
f	–	Darcy friction factor from (8)–(10).
$\sum_i K_i$	–	Total minor-loss coefficient over the wetted portion of the pipe.

10 Computational approach

The coupled ODEs derived above are well suited to numerical integration. A convenient strategy is to treat the filling and draining phases as two separate initial-value problems linked by an event.

- **Filling phase.** Integrate the second-order ODE (12) for $L(t)$ (or, equivalently, the first-order system for $(L, \dot{L})$) starting from the initial conditions (6). At each step, evaluate the container height from $h_c(L) = h_0 - (A/A_0)(L - L_1)$ and update the friction factor f using the instantaneous Reynolds number. Terminate the integration at the event $L(t^*) = L_{\max}$, recording t^* and the front speed $v(t^*) = \dot{L}(t^*)$.
- **Draining phase.** Use $h_c(t^*) = h_0 - \frac{A}{A_0}(L_{\max} - L_1)$ and $v(t^*)$ as initial conditions for the draining system (24)–(26). Integrate forward until either $h_c(t)$ approaches h_{out} or the velocity drops to zero.
- **Loss evaluation.** In both phases, compute the combination $fL/D + \sum_i K_i$ using the current wetted length L (or $L_{\max}$ in draining) and update f from the Churchill formula as the Reynolds number evolves.

Standard ODE solvers such as Runge–Kutta methods with adaptive step size (e.g. those described in Hairer et al. [11]) are adequate.

Example Python implementation

To make the model directly runnable, this subsection provides a complete Python script using `numpy` and `scipy.integrate.solve`$_i vp. It encodes the filling and draining phases, Churchill friction, and event handling fo$

and $h_c = h_{\text{out}}$.

```python
import numpy as np
from math import log
from dataclasses import dataclass
from scipy.integrate import solve_ivp

def churchill_f(Re, eps_over_D):
    Re = max(Re, 1e-8)
    A1 = (-2.457*np.log((7/Re)**0.9 + 0.27*eps_over_D))**16
    A2 = (37530/Re)**16
    return 8 * (((8/Re)**12 + 1.0/((A1 + A2)**1.5))**(1/12))

@dataclass
class Params:
    g: float
    rho: float
    mu: float
    D: float
    A: float
    A0: float
    eps_over_D: float
    Ksum_fill: float
    Ksum_drain: float
    h0: float
    L1: float
    Lmax: float
    h_out: float
    # piecewise linear h(L)
    S: np.ndarray   # breakpoints [S0..S4]
    c: np.ndarray   # intercepts
    s: np.ndarray   # slopes

def h_of_L(L, p: Params):
    # find segment index
    idx = np.searchsorted(p.S, L, side='right') - 1
    idx = np.clip(idx, 0, len(p.S)-2)
    return p.c[idx] + p.s[idx]*L

def filling_rhs(t, y, p: Params):
    L, v = y
    h_c = p.h0 - (p.A/p.A0)*(L - p.L1)
    Re = p.rho*abs(v)*p.D/p.mu
    f = churchill_f(Re, p.eps_over_D)
    losses = f*L/p.D + p.Ksum_fill
    dvdt = (p.g/L)*(h_c - h_of_L(L, p)) - 0.5*(losses/L)*abs(v)*v
    return [v, dvdt]

def event_reach_outlet(t, y, p: Params):
    return y[0] - p.Lmax
event_reach_outlet.terminal = True
event_reach_outlet.direction = 1

def drain_rhs(t, y, p: Params):
```

```
    h_c, v = y
    Re = p.rho*abs(v)*p.D/p.mu
    f = churchill_f(Re, p.eps_over_D)
    losses = f*p.Lmax/p.D + p.Ksum_drain
    dvdt = (p.g/p.Lmax)*(h_c - p.h_out) - 0.5*(losses/p.Lmax)*abs(v)*v
    dhcdt = -(p.A/p.A0)*v
    return [dhcdt, dvdt]

# Example parameters (replace with experiment/geometry)
p = Params(
    g=9.81, rho=1000, mu=1e-3, D=0.01, A=np.pi*(0.01/2)**2,
    A0=0.02, eps_over_D=1e-4, Ksum_fill=1.0, Ksum_drain=1.0,
    h0=0.3, L1=0.05, Lmax=1.2, h_out=-0.4,
    S=np.array([0.0, 0.05, 0.55, 0.95, 1.2]),
    c=np.array([0.0, 0.05, -0.20, 0.10]),
    s=np.array([1.0, 0.5, 0.75, -1.0])
)

# Filling phase
y0 = [p.L1, 0.0]
sol_fill = solve_ivp(
    lambda t, y: filling_rhs(t, y, p),
    [0, 20], y0, rtol=1e-7, atol=1e-9,
    events=lambda t, y: event_reach_outlet(t, y, p)
)
L_end, v_end = sol_fill.y[0, -1], sol_fill.y[1, -1]
t_star = sol_fill.t_events[0][0]

# Draining phase
y0_drain = [p.h0 - (p.A/p.A0)*(p.Lmax - p.L1), v_end]
sol_drain = solve_ivp(
    lambda t, y: drain_rhs(t, y, p),
    [t_star, t_star + 200], y0_drain, rtol=1e-7, atol=1e-9,
    events=lambda t, y: y[0] - p.h_out
)

print(
    "Priming reached outlet at t* = %.4f s, "
    "L = %.3f, v = %.3f m/s" % (t_star, L_end, v_end)
)
if sol_drain.t_events[0].size:
    print(
        "Draining stopped at t = %.2f s when h_c = h_out."
        % sol_drain.t_events[0][0]
    )
```

This implementation mirrors the physical phase change from a partially filled, advancing slug to a fully filled draining siphon and allows direct comparison between simulation and experiment for a given geometry. In practice one would replace the example parameter values by those measured for a specific prototype and, if desired, add diagnostic checks for conditions such as cavitation using the head margin (5).

11 Value, usage, and future work

This document collects the full derivations for a self-starting siphon model and adds several practical elements: (i) explicit start-up conditions matching experimental practice, (ii) an exact container volume map for arbitrary internal pipe layouts together with a constant-area specialisation, (iii) quantitative validity limits for cavitation, capillarity, and non-hydrostatic inlet effects, (iv) an analytic, geometry-only upper bound on the climb height in the idealised lossless, fixed-h_c case, and (v) a runnable Python solver for the two-phase ODE system with Churchill friction and minor losses.

In a design setting the model can be used in two complementary ways: first, to screen candidate geometries via the ideal bound and the geometric priming criterion $h_{\max} > h_3$, and second, to simulate full filling and draining transients for a chosen layout and parameter set. Possible extensions include incorporating a pressure-floor cavitation switch directly into the ODE, adding a dedicated meniscus-start model for very small diameters D, and coupling the slug model to a simple two-phase description of air entrainment and column separation near stall.

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